Injuries in Office Practice

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Definition

The physical damage that results when a human body is suddenly subjected to energy (mechanical, chemical, thermal, and radiation) in amounts that exceeds the threshold of physiological tolerance or else the result of lack of one or more vital elements, such as oxygen.

Classification Injuries

Injuries are classified as:

☑ **Unintentional injuries:**

- Falls
- Fire-related injuries/burns
- Drowning
- Poisoning
- Animal bites

☑ **Intentional injuries:**

- Interpersonal violence (homicide and abuse)
- Self-harm (attempted suicide)

Road traffic accidents (RTAs): A road traffic accident is any injury due to crashes originating from, terminating with or involving a vehicle partially or fully on a public road.

Fall: According to the WHO, a fall is defined as an event which results in a person coming to rest inadvertently on the ground or floor or other lower levels.

Classification Injuries

Burns: It is an injury to the skin or other organic tissue caused by heat or due to radiation, radioactivity, and friction or contact with chemicals. Thermal (heat) burns occur when some, or all of the cells, in the skin or other underlying tissues are damaged/destroyed by hot liquids (scalds), hot solids (contact burns), or flames (flame burns).

Drowning: Drowning is the process of experiencing respiratory impairment from submersion/immersion.

Poisoning: Clinical condition produced by exposure to an agent in doses considered to be toxic.

Management of Common Injuries Encountered in Pediatric Age Group

There are three phases of injury prevention:

1. *Primary prevention:* Preventing new injuries.
2. *Secondary prevention:* Reducing severity of injuries.
3. *Tertiary:* Decreasing the frequency and severity of disability after injury.

General Measures

- ☑ *Immediate resuscitations:* Airway, breathing, and circulation.
- ☑ *Personal protection and infection control:* Use glove, gown, and goggles while examining wound.
- ☑ *Pain and anxiety management:*
  - *Assessment:*
    - Before assessing pain, know the child's level of understanding.
    - Ask the child to use one finger to point to where the pain originates.
    - Pain should be assessed on an ongoing basis, throughout wound management: ideally, before, during, and after a procedure (e.g., dressing removal, cleansing, and dressing application).
    - Use pain assessment scale
  - *Management:*
    - *Goal:* Alleviate or minimize pain and lessen emotional distress
  - *Supportive:*
    - Involve play therapy and distraction (use familiar toys)
    - Breastfeeding or sucrose (neonate/infant)
    - Involve the parent or guardian in care
    - Use a calm and quiet environment
  - *Medication:*
    - Oral analgesics—paracetamol and ibuprofen
    - Opioids and general anesthetic agents—in injuries causing severe pain or for procedure (e.g., exploration of wound)

Head Injury

Assessment of sensorium: Using Glasgow Coma Scale (GCS)

Identify red flag signs:

- ☑ GCS score, including an initial GCS score <14, or for children <1 year, GCS (pediatric) <15 or GCS score <15 at 2 hours after an injury
- ☑ *Head examination:* Tense fontanelles if age <18 months
- ☑ Suspected open, depressed and/or palpable skull fracture
- ☑ Signs of base of skull fracture (periorbital ecchymosis—"panda's eyes", Battle's sign, hemotympanum, cerebrospinal fluid otorrhea, and rhinorrhea)
- ☑ Nonfrontal scalp hematoma, regardless of size
- ☑ New focal neurological deficits after injury
- ☑ For children aged <1 year—presence of bruise, swelling, or laceration of >5 cm on the head, regardless of location

Imaging—CT brain

Mild traumatic brain injury (TBI)—GCS-13-15

- ☑ Observe child 4–6 hours in hospital, if feasible or send home with the advice (explain red flag signs) to observe for 48 hours
- ☑ Follow after 24–48 hours
- ☑ Use telemedicine

Moderate-to-severe TBI

- ☑ Infants and children with severe or moderate TBI (GCS score 3–8 or 9–12, respectively) should receive intensive care unit (ICU) monitoring
- ☑ Look for raised intracranial pressures (ICP) features—vomiting, seizures, irregular breathing, bradycardia, hypotension, and unequal pupil
- ☑ *Follow neuroprotective measures*:
  - Keep head in midline position with 30° elevation
  - Maintain euglycemia, normothermia, and euvolemia
  - Antiseizure medication
  - Antiedema measure (3% NaCl)
- ☑ Shift the child to higher centers with ICU

Specific Management

Cuts and Wounds

Hemostasis: In order to stop bleeding

- ☑ Apply pressure
- ☑ Use tourniquet
- ☑ Elevation of limbs
- ☑ Suturing

Wound cleaning:

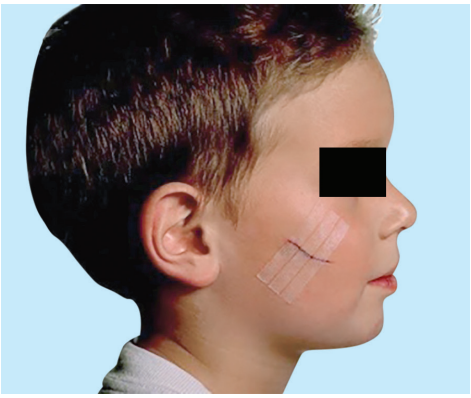
- ☑ Disinfect—clean the surrounding skin with antiseptic (avoid inside the wound)
- ☑ Decontaminate—remove foreign bodies from wound
- ☑ Debridement—remove dead and devitalized tissue
- ☑ Irrigate wound with normal saline
- ☑ Use high pressure irrigation if contamination present
  - High pressure irrigation—irrigate with normal saline with pressure of 5–8 PSI, which is generally achieved with a 35–65 mL syringe attached to a plastic splatter shield, or a 19-gauge needle if a splatter shield is unavailable

- ☑ Administer the irrigation in a pulse-like fashion, pushing the plunger repeatedly, delivering 1–2 mL per squirt
- ☑ At least 100 mL per centimeter of laceration, may need more for grossly contaminated wounds

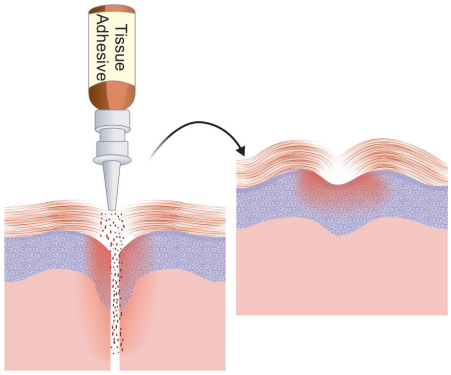
Wound closure:

To hasten wound healing, edges are opposed manually by using following methods:

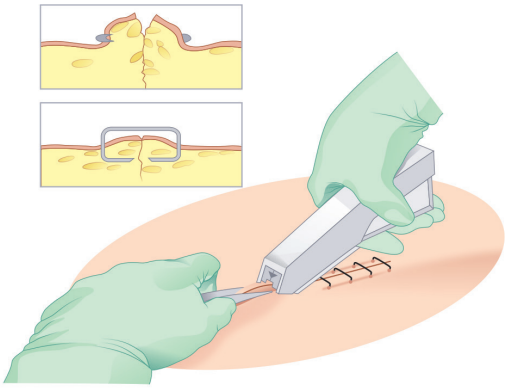
☑ Skin adhesive strip—for small noncontaminated wounds (e.g., steristrip)



☑ Adhesive glue—for small laceration with clear edges



☑ Staples—for small scalp laceration



Specific Management

Suturing:

- ☑ Laceration > 5 cm
- ☑ Deep dermal wounds
- ☑ Locations that are prone to flexion, tension, or wetting

  Suturing should be done under local (lignocaine) or general anesthesia.

Antibiotics: Antibiotics prophylaxis may or may not required

Dressing and follow-up:

- ☑ The first layer should be *nonadherent* (e.g., saline-soaked gauze)
- ☑ Second layer is by *absorbent material* (occlusive dressings with hydrocolloids, hydrogels, and polyurethane films) may be better at accelerating healing, reducing infection, and decreasing pain
- ☑ *Soft gauze tape* to secure the dressing in place

Follow-up:

- ☑ Advise parents to seek medical attention for any signs of infection
- ☑ To keep wound dry
- ☑ Any *sutures or adhesive strips* should be *removed 7–10 days* after initial wound closure (or 3–5 days if on the head); tissue adhesive glue will naturally slough off after 1–2 weeks
- ☑ *Remove dressings* at the same time as the sutures or adhesive strips

Fractures

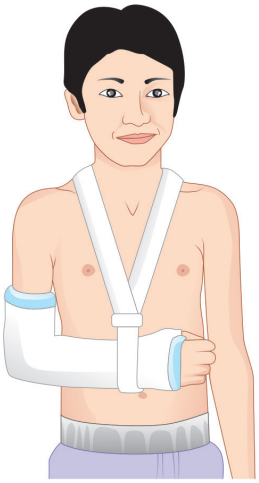
Assess the status of:

- ☑ Skin (bruise open wound)
- ☑ Circulation (color, pulse volume, and CFT)
- ☑ Nerve function (e.g., sensation near thumb web space for radial nerve)

X-Ray of affected limb (Anteroposterior/lateral view)

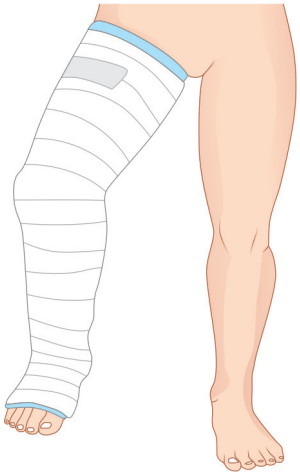
- ☑ Apply splint—specific to fracture site
  - Clavicle fracture—sling or figure of 8 splint

- Humerus fracture and elbow—long arm splint with cuff and collar in gravity mode



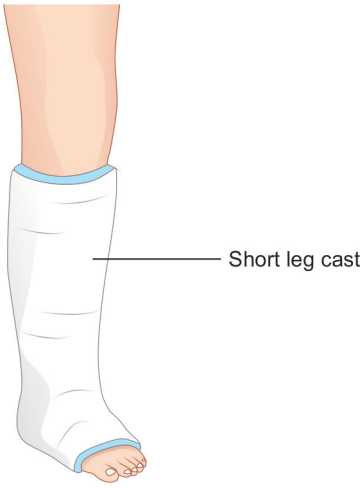
- Forearm—sugar tong splint

- Tibia—long leg splint



Specific Management

- Ankle/foot—short leg splint



- ☑ Pain management (see above text)
- ☑ Wound management
- ☑ Rest
- ☑ Resume normal activity once relived from pain (48 hours)

Burns

Assess area of burn using:
First- and second-degree <10% body surface area (BSA)

- ☑ Silver sulfadiazine ointment; wash the wound before each application
- ☑ AQUACEL Ag dressing (silver impregnated)
- ☑ Change dressing daily
- ☑ Debridement if blisters rupture

Burns with >15% and third-degree burns

- ☑ Fluid resuscitation—according to Parkland formula—4 mL × BSA% × weight in kg (start from the time of injury)
- ☑ Antibiotic—penicillin
- ☑ Shift the child to burn center

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Common Worm Infestations

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